

The Complement Rule Is Excluded Middle: Machine-Checked Probability over Intuitionistic Logic

Anonymous Author(s)

Abstract

Kolmogorov's probability is defined within a classical logic framework, whose complement rule becomes equivalent to the law of excluded middle in the limiting case of certainty. That framework presupposes every event is decidable, which fails for the semidecidable properties, termination, reachability, that verification cares about. We mechanize, in Lean 4 over mathlib, what probability becomes when the event logic is *intuitionistic*: valuations on frames, the algebras of verifiable propositions. Two characterizations anchor the development: (1) the complement rule $v(\neg A) = 1 - v(A)$ ¹ holds for every valuation exactly when the algebra satisfies excluded middle, and (2) Dempster conditioning agrees with Bayesian conditioning at a piece of evidence exactly when that evidence carries zero slack. On finite frames, every valuation is a classical mass function on points. In the general case, a valuation splits into atomic and diffuse parts, and its Boolean core carries a totally monotone Dempster-Shafer belief function. We define the slack of an event A as $1 - v(A) - v(\neg A)$: in the measure model it is the boundary mass. On a semidecidable event, forcing it to zero is exactly as hard as solving the halting problem. We further prove that product valuations are unique on chain products and provably non-unique on the powerset frame, where Fubini fails, show that the naive transcription of Cox's theorem is unsatisfiable, and build the Aczél/Hölder analytic core from scratch.

Keywords: intuitionistic logic, probability, Cox's theorem, belief functions, locales, Lean, mathlib, formalized mathematics

1 Introduction

Classical logic underlies Kolmogorov's probability [Kolmogorov 1933]. The axioms of this probability specify an arithmetic of chance (non-negativity, normalization, σ -additivity). Beneath that arithmetic sits a logic: the events form a Boolean σ -algebra, closed under complement and satisfying excluded middle. The complement rule, $P(A^c) = 1 - P(A)$, is where the two layers touch. It is an arithmetical equation about a logical operation, obeyed by every assignment of degrees of certainty. In the limiting case of all-or-nothing probabilities, it becomes equivalent to the law of excluded middle.

¹By $\neg a$ we mean the pseudocomplement of a in a frame.

That framework commits to logical axioms that computer science cannot always commit to. Many properties of programs (does this computation halt, does this process reach a given state) are only *semidecidable*: a procedure confirms the property when it holds, and no procedure, in general, confirms its failure. The natural event algebra for a semidecidable property is a *frame*, the algebra of open sets of a space, and equivalently the algebraic semantics of intuitionistic logic. We show that a classical, fully resolved (slack-free) probabilistic reading of a semidecidable property is exactly as hard as solving the halting problem. This reading is not exotic: on the space of infinite binary tapes, the value of the halting event is Chaitin's Ω [Chaitin 1975], and its slack is the mass of the topological boundary of the halting event. Reasoning about the chance that a program has some property therefore needs an event logic that is not Boolean. We align the probability calculus with the logic of the objects it reasons about, and mechanize the result.

Probability is, as the Cox-Jaynes tradition has it, *extended logic* [Cox 1946; Jaynes 2003]. A probability theory has two layers. One layer is the event algebra: the underlying set of events, and how they combine under the algebraic operations of conjunction, disjunction, and negation. The other layer is the arithmetic: axioms on maps from the event algebra to numbers, saying which assignments of degrees of certainty to the events are admissible. Kolmogorov takes the event algebra to be Boolean, and his axioms tie the arithmetic to that choice, most visibly in the complement rule. The logic of events is in this sense a *parameter* of the theory, and Kolmogorov's calculus is its value at Boolean logic. Quantum probability, which re-bases the arithmetic on orthomodular lattices, is the established precedent for varying it [Birkhoff and von Neumann 1936; Gleason 1957]. We compute the value at intuitionistic logic, whose event algebras are frames: complete Heyting algebras, the lattices of open sets. Two facts make the result a theory of *probability* rather than an arbitrary grading of events. In the limit of certainty it returns the logic: the $\{0, 1\}$ -valued valuations are exactly the prime-ideal points of the frame (Section 3). And at Boolean logic it returns the classical theory: complete Boolean algebras are exactly the frames satisfying excluded middle, and every valuation there obeys the classical complement rule. So, the classical theory embeds as a special case.

The theory that results fixes the arithmetic on the event algebra and imposes no axiom on complements. A *valuation* on a frame Ω is a monotone, normalized, modular map $v : \Omega \rightarrow [0, \infty]$, i.e. such that $v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b)$.

The pseudocomplement $\neg a$ satisfies $a \sqcap \neg a = \perp$ but not, in general, $a \sqcup \neg a = \top$, so the complement rule degrades to $v(a) + v(\neg a) \leq 1$, with a gap

$$\text{slack}(a) = 1 - v(a) - v(\neg a).$$

Classically, this is the width of the Dempster–Shafer belief/plausibility interval [Dempster 1967; Shafer 1976], quantifying ignorance. In our system it instead quantifies the mass of the part of a that is neither confirmed nor refuted, the probabilistic image of the semidecidable gap above. On the opens of a space with measure μ , the slack of an open U is $\mu(\partial U)$, the mass of its topological boundary. For verification this matters concretely: knowing how likely a property is to hold no longer tells you how likely it is to fail, and the gap between the two, the slack, is now also required to answer that question.

Slack is necessarily submodular, a consequence of v 's own modularity. It does not, however, determine v : two fully modular valuations on the same frame can share one slack function, a fact we exhibit on an explicit three-point space. Weak excluded middle, $\neg\neg a \sqcup \neg a = \top$, does not repair this. Imposed on the whole frame it collapses slack to the double-negation gap alone, and the fully Boolean case, where even that gap vanishes, is the worst case for recovery: slack is identically zero there for every valuation.

The conditional $v(a \sqcap b)/v(b)$, the product rule, and Bayes symmetry hold verbatim in the resulting theory, which reduces to intuitionistic logic in the limit of certainty. Certain identities vary, however. The classical denominator of Bayes' theorem, $P(B) = P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A)$, degrades to a lower bound whose deficit is a conditional slack. Marginalization is exact only over partitions. The complement trick $P(A) = 1 - P(\neg A)$ degrades to the interval $[v(A), 1 - v(\neg A)]$ of width $\text{slack}(A)$. Dempster's rule and Bayesian conditioning, classically indistinguishable, give different posteriors from the same evidence unless its slack vanishes. The value of an event A itself is read as $P(\text{int } A)$, so a nonatomic distribution queried on intervals still behaves classically: an interval's boundary is two points, carrying no mass. Differences arise only at boundaries carrying mass and at semidecidable hypotheses.

Cox's original theorem required repair after Halpern's counterexample [Cox 1946; Halpern 1999; Horn 2003]. Even with this repair, the direct translation of Cox's theorem to intuitionistic logic is *unsatisfiable*. This is because the sum rule is irreducible: a disjunction's value is provably not a function of its disjuncts' values, so modularity must be posited rather than derived. Weak excluded middle, an assumption strictly between intuitionistic and classical logic, does not restore the reducibility of the sum rule: the counterexample frame itself satisfies weak excluded middle.

Contributions.

1. **Two characterizations** (Sections 3 and 5). The complement rule holds for *every* valuation iff the frame satisfies excluded middle, the exact content of Van Horn's classical negation axiom R3 [Horn 2003]. Dempster and Bayesian conditioning agree iff the evidence has zero slack.
2. **Structure theorems** (Sections 4 and 5). On finite frames a valuation is a classical mass function on points. In general it splits into atomic and diffuse parts, and on its Boolean core it is a belief function.
3. **Cox's theorem, dissected** (Section 6, Appendix C). The product-rule half is independent of the logic, proved by an Aczél/Hölder generator built from scratch. The sum-rule half, a theorem classically, admits no derivation here (a five-element counterexample) and must be posited.
4. **Two grounding models** (Section 7). Any classical measure, read on the opens, is a valuation whose slack is boundary mass. A slack-free readout of halting is uncomputable, and on Cantor space the value of the halting event is Chaitin's Ω .
5. **The monad dichotomy** (Section 9). Product valuations are unique on chain products and provably non-unique on the powerset frame, where Fubini fails. The countable fragment composes associatively and commutatively.
6. **Weak excluded middle** (Sections 2 and 7). Global weak excluded middle on the opens of X is exactly extremal disconnectedness of X , for compact Hausdorff X exactly the Stonean spaces.
7. **A first formal treatment**. Everything above is machine-checked in Lean 4 over mathlib, to our knowledge the first mechanization of non-additive probability over intuitionistic logic.

Mechanization. The development is carried out in Lean 4 [de Moura and Ullrich 2021] over mathlib [mathlib Community 2020], in roughly 5,000 lines with no sorry, and is included in full as anonymous supplementary material.² Two points are specific to a proof assistant. We separate *object-level* classicality, excluded middle in the frame Ω , from the *meta-level* classicality mathlib inherits through `Classical.choice`, track which results use which, and equip the one essential non-Boolean model with a computable decidability instance. And the product-rule half of Cox's theorem needs an Aczél/Hölder regratuation absent from mathlib, which we build from scratch.

²Claude (Anthropic) was used to assist with drafting and debugging the Lean formalization.

2 Valuations on Frames

2.1 The Event Logic

Definition 1 (frame). A *frame* is a complete lattice Ω in which finite meets distribute over arbitrary joins,

$$a \sqcap \left(\bigsqcup_{i \in I} b_i \right) = \bigsqcup_{i \in I} (a \sqcap b_i)$$

for every $a \in \Omega$ and every family $\{b_i\}_{i \in I} \subseteq \Omega$. Equivalently, it is a complete Heyting algebra. By $x \Rightarrow y$ we denote the Heyting implication in a frame, and $\neg a := a \Rightarrow \perp$ is the pseudocomplement of $a \in \Omega$. Also, a^c denotes a complement of a , if it exists, i.e. an element such that $a \sqcap a^c = \perp$ and $a \sqcup a^c = \top$.

Frames are the algebraic semantics of intuitionistic logic and the point-free abstraction of topological spaces. The same objects with morphisms reversed form the category of *locales*, so the distinction is invisible at the level of a single algebra. We say *frame* throughout and reserve *locale* for the spatial and categorical statements (points of a locale, classifying locales) where that word is the standard one. The motivating example throughout is $\Omega = \mathcal{O}(X)$, the opens of a space, with $\sqcap, \sqcup$ as intersection and union and the pseudocomplement $\neg a$ of an open given by the *interior* of its set complement. In mathlib this is the typeclass `Order.Frame`, and the Heyting operations come with the key asymmetry already in place: $a \sqcap \neg a = \perp$ holds in any Heyting algebra (`inf_compl_eq_bot`), while $a \sqcup \neg a = \top$ is available only in Boolean algebras. An element with $a \sqcup \neg a = \top$ is *complemented*: it satisfies its own instance of excluded middle.

2.2 The Valuation Structure

The central object assigns a degree of certainty to each element of the frame, with inclusion–exclusion as the only interaction axiom:

Definition 2 (valuation). A *valuation* on a frame Ω is a *monotone* function $v : \Omega \rightarrow [0, \infty]$ with $v(\perp) = 0$, $v(\top) = 1$ (*normalization*), and, for all $a, b \in \Omega$, *modularity*:

$$v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b).$$

```
structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b
    = toFun (a ⊔ b) + toFun (a ⊓ b)
```

This axiom set is not ours: Weatherson’s axioms for intuitionistic probability are exactly these [Weatherson 2003], and the valuations of constructive measure theory [Coquand

and Spitters 2009; Vickers 2008] are their Scott-continuous instances (Section 8).

Values live in the extended non-negative reals $[0, \infty]$, and normalization confines them to $[0, 1]^{\text{Lean}}$. The conventions this choice carries, unconditional summation and truncated subtraction, are an artifact of the mechanization and are recorded in Appendix A.

2.3 Slack

Modularity applied to the disjoint pair $(a, \neg a)$ collapses to $v(a) + v(\neg a) = v(a \sqcup \neg a)^{\text{Lean}}$. So, the join need not be $\top$:

Theorem 3 (the complement bound). *For every valuation v on a frame Ω and every $a \in \Omega$, $v(a) + v(\neg a) \leq 1$.*

```
theorem Valuation.add_compl_le_one (v :
  Valuation Ω) (a : Ω) :
  v a + v a^c ≤ 1
```

This is a *derived* Dempster–Shafer sub-additivity inequality. The deficit

$$\text{slack}(a) = 1 - (v(a) + v(\neg a)) = 1 - v(a \sqcup \neg a)$$

is the valuation of the undecided region. It decomposes canonically. The pair $(\neg\neg a, \neg a)$, double negation against negation, is always disjoint. However, it is a partition only under *weak* excluded middle ($\neg a \sqcup \neg\neg a = \top$). Applying modularity to this disjoint pair splits the slack into two logically independent gaps:

Definition 4 (regular element). An element $a \in \Omega$ is *regular* (or $\neg\neg$ -stable) if $a = \neg\neg a$.

Theorem 5 (slack decomposition^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{slack}(a) = \text{dnGap}(a) + \text{deMorganGap}(a)$, where $\text{dnGap}(a) = v(\neg\neg a) - v(a)$ is 0 if a is $\neg\neg$ -stable (regular), and $\text{deMorganGap}(a) = 1 - v(\neg\neg a \sqcup \neg a)$ is 0 if weak excluded middle holds at a .*

The single Dempster–Shafer “ignorance” number is thus resolved into a regularity defect and a De Morgan defect: the first measures failure of double-negation stability, the second corresponds to a named intermediate logic (weak excluded middle). Constructively, the contrapositive of a universal hypothesis returns only its double-negated form, so for open sets, this specification of probability does not fall victim to Hempel’s raven paradox [Hempel 1945]: evidence of the white sock indoors confirms at most $\neg\neg a$, not a itself, and dnGap is exactly the gap between confirming $\neg\neg a$ and confirming a itself.

Because dnGap and deMorganGap are themselves defined from v , the apparent three unknowns $v(a)$, $v(\neg a)$, and $v(\neg\neg a)$ are not independent: they are part of a three-constraint system, yielding the following result:

Theorem 6 (slack in closed form^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$,*

$$\text{slack}(a) = (v(\neg\neg a) - v(a)) + (1 - v(\neg\neg a) - v(\neg a)),$$

where $v(a) \leq v(\neg\neg a)$.

The inequality follows from monotonicity applied to $a \leq \neg\neg a$, a fact of every Heyting algebra, and shows the first summand is exactly the difference, not the floor produced by $[0, \infty]$'s truncated subtraction. The split into a regularity defect and a De Morgan defect follows from monotonicity and modularity alone, with no further axiom about ignorance required. This contrasts with the usual Dempster–Shafer picture, where the mass assigned to non-singleton, ambiguous sets is an extra primitive.

Theorem 6 expresses slack pointwise in terms of v . The converse, whether slack in turn determines v , is more delicate. One constraint is automatic:

Theorem 7 (slack is submodular^{Lean}). *For every valuation v on a frame Ω and every $a, b \in \Omega$,*

$$\text{slack}(a \sqcup b) + \text{slack}(a \sqcap b) \leq \text{slack}(a) + \text{slack}(b).$$

The proof runs v 's own modularity twice, once at (a, b) and once at $(\neg a, \neg b)$, and combines the two results with the De Morgan laws: the equality $\neg(a \sqcup b) = \neg a \sqcap \neg b$, which holds in every Heyting algebra, and the inequality $\neg a \sqcup \neg b \leq \neg(a \sqcap b)$, monotone under v but not in general an equality. Any function proposed as the slack of some valuation must therefore be submodular.

Remark 8 (non-recovery^{Lean}). Submodularity is necessary, not sufficient: Section 7.2 exhibits two valuations on the same frame with identical slack functions.

Remark 9 (slack under weak excluded middle^{Lean}). The two gaps, dnGap and deMorganGap , computed at a single element, are logically independent. However, if weak excluded middle is imposed on the frame for every a ($\neg\neg a \sqcup \neg a = \top$ ^{Lean}), then deMorganGap vanishes identically and $\text{slack}(a) = \text{dnGap}(a)$ for every valuation and every a . This sits strictly between plain intuitionistic logic, where both gaps can be positive, and full excluded middle, where both vanish^{Lean}. A concrete instance appears in Section 7.

2.4 What Survives without Excluded Middle

Three classical pillars survive intact, and one fails in a precisely delimited way. *Disjoint additivity* is unconditional: $a \sqcap b = \perp$ implies $v(a \sqcup b) = v(a) + v(b)$. *Conditioning* is native: for $v(b) \neq 0$ the posterior $v(a \mid b) = v(a \sqcap b) / v(b)$ is again a valuation (the proof is frame distributivity), and the product rule $v(a \mid b) \cdot v(b) = v(a \sqcap b)$ and Bayes symmetry (and therefore Bayesian updating) $v(a \mid b) v(b) = v(b \mid a) v(a)$ hold^{Lean}. *Total probability* holds over any genuine partition, meaning $a \sqcap a' = \perp$ and $a \sqcup a' = \top$: $v(b) = v(a \sqcap b) + v(a' \sqcap b)$. What fails is total probability over $\{a, \neg a\}$. That family need not join to $\top$, and the conditional masses fall short of $v(b)$ by a conditional slack^{Lean}.

The classical fragment. We characterize the elements of Ω on which the classical complement rule holds. The $\neg\neg$ -stable (*regular*) elements, which form a Boolean algebra (by Glivenko's theorem), are a natural candidate, but they do not suffice. Modularity constrains the frame join $\sqcup$, whereas the Boolean join of regular elements is the regularized $\neg\neg(a \sqcup b)$, for which only the inequality $v(a) + v(b) \leq v(\neg\neg(a \sqcup b)) + v(a \sqcap b)$ survives (i.e. the 2-monotonicity of Section 5). A regular but uncomplemented element, which is an open ray $(-\infty, 0)$ in the opens of $\mathbb{R}$, under an atomic measure at 0, still carries slack. The correct classical fragment is the *complemented* elements, where $v(a) + v(a^c) = 1$ holds exactly^{Lean}. On a connected locale that fragment can be as small as $\{\perp, \top\}$. In the Boolean limit, every element is complemented, the slack vanishes identically, and Kolmogorov is recovered^{Lean}.

Mixtures. Valuations are closed under finite convex combination^{Lean}, with every axiom inherited termwise, because modularity is linear. This innocuous fact becomes a characterization in Section 4.

3 Classicality in Theory and Metatheory

Classicality enters this work twice: in the metatheory, the logic of the proof assistant itself, and in the object theory, the frame's own logic.

Lean's meta-theory. Lean's foundations include `Classical.choice`, and `mathlib` is built on it throughout. For example, the order relation on $\mathbb{R}$ and $[0, \infty]$ has no computable Decidable instance in general, so any library lemma that case-splits on $x \leq y$ for reals resolves that split via `Classical.propDecidable`. A formalization that never invokes excluded middle at the level of frame elements $a \in \Omega$ can therefore still be classical through the arithmetic it borrows from the library. We track this separately from object-level classicality throughout the paper. Theorem 20 of Section 4 rests only on this ambient baseline and contains no explicit case split. The Scott-continuity recovery theorem (Theorem 19) invokes Lean's classical tactic directly, and the explicit Cox model of Section 6 goes the other way, replacing the default with an explicit computable Decidable instance to keep that one instance genuinely non-classical.

Object-level classicality: the R3 axiom. Van Horn's repaired Cox theorem [Horn 2003] axiomatizes plausibility over classical propositions. Its negation axiom, called *R3*, posits that the plausibility of $\neg A$ is a fixed non-increasing function S of the plausibility of A . Consistency forces the involution $S \circ S = \text{id}$, and therefore the complement rule $p \mapsto 1 - p$. We show that the involutive negation is double-negation elimination, so *R3* is where classical logic enters

every known Cox-style derivation. To do this, we first isolate the exact content of R3's conclusion as a predicate on valuations, then prove equivalence to excluded middle.

That predicate is *classical negation*: a valuation v has classical negation if $v(a) + v(\neg a) = 1$ for every $a \in \Omega$.

Theorem 10 (the R3 characterization^{Lean}). *For every frame Ω , every valuation on Ω has classical negation iff $a \sqcup \neg a = \top$ for every $a \in \Omega$.*

The forward direction is a calculation. The content is the converse: wherever excluded middle fails, a valuation with genuine slack must be *constructed*. Suppose $a \sqcup \neg a \neq \top$.

Definition 11 (ideal, prime ideal). An ideal $J \subseteq \Omega$ is downward closed and closed under finite joins. J is *prime* if $\top \notin J$ and $a \sqcap b \in J$ implies $a \in J$ or $b \in J$.

The prime ideal separation theorem (mathlib's `DistribLattice.prime_ideal_of_disjoint_filter_ideal`) yields a prime ideal J containing $a \sqcup \neg a$ but not $\top$. The indicator of J 's complement,

$$v(x) = \begin{cases} 0 & x \in J \\ 1 & x \notin J, \end{cases}$$

is a valuation. Modularity in the only nontrivial case ($x, y \notin J$) is *exactly* primeness, $x \sqcap y \in J \Rightarrow x \in J \vee y \in J$. And this valuation assigns $v(a) + v(\neg a) = 0 \neq 1$.

Sharp valuations are the points of the locale. The construction above is one half of an exact correspondence. Call v *sharp* if it is $\{0, 1\}$ -valued, a state of full certainty. Sharpness characterizes exactly the prime-ideal indicators:

Theorem 12 (sharp valuations are points^{Lean}). *For every valuation v on a frame Ω , v is sharp iff v is the complement-indicator of a prime ideal.*

The correspondence is given by two mutually inverse maps: $v \mapsto Z(v) = \{x \in \Omega : v(x) = 0\}$, which computes the zero-set of a sharp valuation, and the map taking J to its complement-indicator (the valuation of Theorem 10's construction). $Z(v)$ is prime exactly because v is sharp, and the complement-indicator of $Z(v)$ recovers v itself. In locale-theoretic terms, sharp valuations recover the *points* of the locale. A prime ideal is *completely prime* when it is also closed under arbitrary (not just finite) suprema of its members. The finitely prime points of Theorem 12 sharpen to the completely prime points exactly where Scott continuity is added, recovering points of the locale in the sense used by [Coquand and Spitters 2009; Vickers 2008]:

Theorem 13 (Scott-continuous sharp valuations^{Lean}). *For every valuation v on a frame Ω , v is sharp and Scott-continuous iff v is the complement-indicator of a completely prime ideal.*

The characterization and the two correspondences together locate the object-level classicality precisely: whenever excluded middle fails at some a , the failure occurs at a point of the locale.

4 Valuation Representation

Every valuation on a finite frame is a probability mass function on points. On infinite frames, this can fail, and Scott continuity marks the boundary between where it still holds and where it does not.

4.1 Finite Frames Representation

Theorem 14 (Birkhoff duality [Birkhoff 1937]). *Every finite distributive lattice is isomorphic to the lattice of lower sets of a finite poset: its sup-irreducible elements form a poset P whose lower sets reconstruct the lattice, and every finite poset arises this way from its own lattice of lower sets.*

Let P be the finite poset of sup-irreducible elements of a finite frame. Then, by Birkhoff duality (`OrderIso.lowerSetSupIrred`), every finite frame is the lattice of lower sets of P . We fix the following notation, used throughout the rest of the paper. A *lower set* $L \subseteq P$ satisfies $p \in L, q \leq p \Rightarrow q \in L$, and an *upper set* is defined dually. Lower sets are always denoted L , and upper sets U (opens of a topological space are still U, V, W , as in traditional usage). The *principal lower set* at p is $\downarrow p = \{q : q \leq p\}$, the *strict principal lower set* at p is $\Downarrow p = \{q : q < p\}$, and the *principal upper set* at p is $\uparrow p = \{q : q \geq p\}$. We prove the finite representation theorem below on the lower sets of P . Transporting it across the isomorphism above then gives the theorem for every finite frame, not just lower sets^{Lean}. Let us define the *mass* of a point as the discrete derivative of v^{Lean} :

$$\text{mass}(p) = v(\downarrow p) - v(\Downarrow p).$$

The masses recover v and sum to 1:

Theorem 15 (finite representation^{Lean}). *For a finite P , any valuation v on $\text{Low}(P)$, and any $L \in \text{Low}(P)$, $v(L) = \sum_{p \in L} \text{mass}(p)$, and $\sum_p \text{mass}(p) = 1$.*

The proof works by induction on $|L|$: repeatedly remove a maximal point p from L , and use modularity applied to $(L \setminus \uparrow p, \downarrow p)$ to show each removal changes v by exactly $\text{mass}(p)$. The mass function is a classical probability mass function on P^{Lean} . And since lower sets are exactly the opens of the Alexandrov topology, where interior is the identity, the theorem also reads $v = \mu \circ \text{int}$, i.e. every valuation on a finite frame is the classical measure μ applied after taking the interior. In the finite case, this is the converse of the measure-theoretic bridge of Section 7. The intuitionistic mass also agrees with the classical notion of the mass of a point: reading $v = \mu \circ \text{int}$,

$$\text{mass}(p) = \mu(\downarrow p) - \mu(\Downarrow p) = \mu(\{p\}),$$

since $\downarrow p$ is the disjoint union of $\downarrow p$ and $\{p\}$ and μ is defined on all subsets of P .

Theorem 15 represented v as a sum of point masses. The dual view represents v as a mixture of sharp *point-valuations*. For each $p \in P$, let δ_p be the valuation with $\delta_p(L) = 1$ if $p \in L$, and $\delta_p(L) = 0$ otherwise:

Theorem 16 (mixture characterization^{Lean}). *On a finite frame, every valuation v satisfies $v = \sum_p \text{mass}(p) \cdot \delta_p$.*

Together with the fact that any mixture of valuations is again a valuation^{Lean}, this theorem characterizes the valuations on a finite frame exactly: they are the finite convex mixtures of point-valuations, closed under mixing because modularity is linear. Section 6 shows modularity cannot instead be *derived* from more primitive axioms. It must be posited outright, and doing so is exactly what makes every valuation a mixture of sharp, all-or-nothing certainties.

4.2 Infinite Boundary Representation via Scott Continuity

Both halves of the finite theorem fail for infinite frames:

Example 17 (obstruction^{Lean}). On $\text{Low}(\mathbb{N})$, the indicator v of $\top$ is a valuation with $\text{mass}(p) = 0$ for every point p , and $v(\top) = 1$.

No point carries any of the mass. The failure is one of approximation: $v(\top)$ is not the supremum of $v(\downarrow n)$ over the finite stages. In order to recover the representation, we introduce Scott continuity:

Definition 18 (directed set, Scott continuity). A subset $D \subseteq \Omega$ is *directed* if, for any finite subset $F \subseteq D$, there exists an element $d \in D$ that is an upper bound of F (taking $F = \emptyset$ shows a directed set is nonempty). A valuation v is *Scott-continuous* if $v(\bigsqcup D) = \sup_{a \in D} v(a)$ for every directed D .

The $\top$ -indicator of Example 17 is not Scott-continuous^{Lean}, and Scott continuity is exactly what restores the representation:

Theorem 19 (recovery^{Lean}). *Let v be a valuation on $\text{Low}(\mathbb{N})$. If $v(\top) = \sup_n v(\downarrow n)$ then $v(\top) = \sum_n \text{mass}(n)$.*

In general, on any poset all of whose principal lower sets are finite, Scott continuity implies zero diffuse part without additional preconditions. However, subtraction in $[0, \infty]$ is truncated at 0, so if $\sum_p \text{mass}(p)$ were greater than $v(\top)$, the expression $v(\top) - \sum_p \text{mass}(p)$ would still evaluate to 0. Theorem 20 rules out $\sum_p \text{mass}(p) > v(\top)$ unconditionally, so the subtraction below is never truncated:

Theorem 20 (general decomposition^{Lean}). *For every valuation v on $\text{Low}(P)$ and an arbitrary poset P , $\sum_p \text{mass}(p) \leq v(\top)$.*

By Theorem 20 the remainder $v(\top) - \sum_p \text{mass}(p)$ is a genuine difference. Every valuation thus splits unconditionally

into an *atomic* part carried by points ($\sum_p \text{mass}(p)$, which is always a sub-probability), and a *diffuse* (nonnegative) remainder $v(\top) - \sum_p \text{mass}(p)$. This is a constructive Lebesgue-style decomposition. The finite and Scott cases are the purely atomic extremes, and the $\top$ -indicator is the purely diffuse one. The proof of Theorem 20 re-runs the maximal-point induction on each finite subset of P , bounding its partial sum by $v(\top)$, then takes the supremum over all finite subsets to bound $\sum_p \text{mass}(p)$ itself. Its proof does not directly invoke an excluded-middle case split. However, it relies on the (classical) decidability of the ordering of the reals (a default in mathlib). Representing the diffuse part on non-spatial frames remains the open frontier. The obstruction there is again that the frame has *too few points*, the same one as behind the undecidability of Section 7.

5 The Dempster–Shafer Reading

Sections 2–4 independently arrive at the same structure as Dempster–Shafer (DS) theory [Dempster 1967; Shafer 1976], including sub-additive complements, a belief/plausibility interval, and mass functions. This section makes the identification exact in both statics and dynamics. In each case, it locates the precise point where excluded middle makes a difference.

5.1 Total Monotonicity via Constructive Inclusion–Exclusion

The $\neg\neg$ -stable (*regular*) elements of a frame form a Boolean algebra with meet $\sqcap$ and join $\neg\neg(a \sqcup b)$ (Glivenko). Let us call the resulting algebra the *Booleanization* of this set.

Definition 21 (belief function, plausibility function). Let B be a Boolean algebra. A DS *belief function* on B is a monotone map $\text{Bel} : B \rightarrow [0, 1]$ with $\text{Bel}(\perp) = 0$ and $\text{Bel}(\top) = 1$ that is *n-monotone* for every $n \geq 2$: for all $A_1, \dots, A_n \in B$,

$$\text{Bel}\left(\bigvee_{i=1}^n A_i\right) \geq \sum_{\emptyset \neq T \subseteq \{1, \dots, n\}} (-1)^{|T|+1} \text{Bel}\left(\bigwedge_{i \in T} A_i\right).$$

Its *plausibility function* is the map $\text{Pl} : B \rightarrow [0, 1]$ given by $\text{Pl}(A) = 1 - \text{Bel}(\neg A)$.

We show v , restricted to its Booleanization, is such a belief function. The 2-monotone case is immediate from modularity^{Lean}. The full tower follows from the following result:

Theorem 22 (inclusion–exclusion^{Lean}). *For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$, inclusion–exclusion holds with equality for the frame join:*

$$v\left(\bigsqcup_{i \in s} a_i\right) = \sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right).$$

We state this in Lean additively, with the odd-cardinality half of the signed sum set against the even-cardinality half, so

that no $[0, \infty]$ subtraction occurs. The proof is by induction with modularity and frame distributivity.

Replacing the frame join $\sqcup_i a_i$ on the equality's other side with its double negation $\neg\neg(\sqcup_i a_i)$, the Booleanization's own join, yields the following inequality:

Theorem 23 (total monotonicity^{Lean}). *For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$:*

$$\sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigcap_{i \in T} a_i\right) \leq v\left(\neg\neg\left(\bigcup_{i \in s} a_i\right)\right).$$

The Booleanization's join is given by $\neg\neg \circ \sqcup$, and its meets of regular elements are again regular. Interpreted in this way, Theorem 23 states exactly the ∞ -monotone tower: v restricted to its Booleanization is a belief function.

The proof of Theorem 23 from Theorem 22 follows from the monotonicity along $a \leq \neg\neg a$. Here, inclusion-exclusion does *not* fail constructively. It is exact for the intuitionistic disjunction. Applying the double-negation nucleus (a closure operator preserving finite meets) $\neg\neg$ to the join on the equality's right-hand side can only enlarge it. This follows from the fact that $a \leq \neg\neg a$ and v are monotone, which turns the equality into the inequality of Theorem 23. The amount by which the right-hand side is enlarged is exactly the double-negation gap of Section 2.

Definition 21 is the classical notion, stated on an abstract Boolean algebra. In our setting each of its ingredients is already present: the Boolean algebra is the Booleanization, the belief function is v itself, restricted to the regular elements and evaluated against the Boolean join $\neg\neg \circ \sqcup$, and the plausibility function is $\text{Pl}(a) = 1 - v(\neg a)$. Additive probability does not become a belief function by weakening an axiom, since modularity for v is never relaxed. What changes is only the join used to state the tower. The belief-plausibility interval is governed by the slack:

Theorem 24 (interval width^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{Pl}(a) - v(a) = \text{slack}(a)$.*

5.2 The Conditioning Characterization

DS theory carries two update rules that coincide classically. *Geometric* conditioning renormalizes belief, and under the DS dictionary it is the localic posterior $v(a \sqcap b)/v(b)$ of Section 2.

Definition 25 (Dempster's rule of conditioning^{Lean}). *For a valuation v on a frame Ω , elements $a, b \in \Omega$, and evidence of nonzero plausibility, $v(b) \neq 0$, Dempster's rule of conditioning [Shafer 1976] states that*

$$v(a \parallel b) = \frac{v(a \sqcup \neg b) - v(\neg b)}{1 - v(\neg b)}.$$

The hypothesis $v(b) \neq 0$ makes the denominator nonzero, since $v(b) + v(\neg b) \leq 1$ gives $v(b) \leq 1 - v(\neg b)$.

That is, the DS conditioning rule discards the mass incompatible with the evidence and renormalizes by its plausibility. Both the geometric posterior of Section 2 and Dempster's rule of Definition 25 are again valuations. Their modularity is the frame distributivity $(a \sqcup \neg b) \sqcap (a' \sqcup \neg b) = (a \sqcap a') \sqcup \neg b$. Their disagreement is given by the *excluded-middle gap* $\text{emGap}(a, b) = v(a) - v(a \sqcap (b \sqcup \neg b))$, i.e. the mass of a outside b 's instance of excluded middle:

Lemma 26 (the join-complement identity^{Lean}). *For every valuation v on every frame Ω and every $a, b \in \Omega$, $v(a \sqcup \neg b) = v(\neg b) + v(a \sqcap b) + \text{emGap}(a, b)$.*

Dempster's numerator therefore exceeds Bayes's by exactly $\text{emGap}(a, b)$. This gap depends on a , but whether it vanishes at every a depends only on the slack at the evidence b :

Theorem 27 (the conditioning characterization^{Lean}). *For every valuation v on every frame Ω and every $b \in \Omega$ with $v(b) \neq 0$, $(v(a \parallel b) = v(a \sqcap b)/v(b) \text{ for every } a \in \Omega) \text{ iff } \text{slack}(b) = 0$.*

The hypothesis $\text{slack}(b) = 0$ eliminates the gap $\text{emGap}(a, b)$ for every a and makes the two normalizers coincide, since $1 - v(\neg b) = v(b)$. Conversely, testing at $a := b$, geometric conditioning gives certainty 1, while Dempster gives $v(b)/(1 - v(\neg b))$, which is 1 only if the slack vanishes. Theorem 27 is the valuation-relative analogue of Theorem 10: both are iff statements about the same proposition, $b \sqcup \neg b = \top$.

We call the frame-level fact *static* and the valuation-level fact *dynamic*: the former is an equation in the frame itself, between $b \sqcup \neg b$ and $\top$, with no valuation involved, while the latter tests the same proposition through v , since $\text{slack}(b) = 0$ is exactly $v(b \sqcup \neg b) = v(\top)$. Dempster and Bayes therefore coincide at b whenever v cannot distinguish $b \sqcup \neg b$ from $\top$, even if the two differ in the frame itself. In the measure model, Dempster and Bayes disagree precisely when the evidence's boundary carries mass.

6 Cox's Theorem

Cox's theorem [Cox 1946; Jaynes 2003] derives the probability calculus from conditions on plausibility. After Halpern's counterexample [Halpern 1999], its repaired forms [Arnborg and Sjödin 2000; Horn 2003] fix the calculus up to monotone regratuation. Every known derivation is classical, and Section 3 showed why: the negation axiom R3 is excluded middle. Dropping R3 leaves the product rule unaffected, since it is logic-free, and requires the sum rule to become an axiom rather than a theorem. Separately, the naive constructive statement, demanding a regratuation strictly monotone on all of $\mathbb{R}$, is unsatisfiable (Section 6.3).

6.1 A Cox Structure without Negation

Definition 28 (Cox model^{Lean}). A Cox model on a frame Ω is a plausibility assignment $\text{pl} : \Omega \times \Omega \rightarrow \mathbb{R}$, written $\text{pl}(a, b)$ for the plausibility of a given b , together with a conjunction functional $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, such that $\text{pl}(\cdot, c)$ is monotone for every c , the boundary conditions $\text{pl}(\perp, c) = 0$ and $\text{pl}(\top, c) = 1$ hold for every c , the product rule $\text{pl}(a \sqcap b, c) = F(\text{pl}(a, b \sqcap c), \text{pl}(b, c))$ holds, F is associative, and $x \mapsto F(x, y)$ is strictly monotone for every $y > 0$.

The missing R3 axiom can be observed in the omission of any constraint on the relation between $\text{pl}(\neg a, b)$ and $\text{pl}(a, b)$. The definition restricts the monotonicity of F to $0 < y$ to avoid a contradiction with the fact that the boundary axioms force $F(x, 0) = 0$ for all x . An explicit model on the non-classical chain $[0, \infty]$ satisfies this constraint^{Lean}, as $a = \top$ is provably decidable.

6.2 The Sum Rule Is Irreducible

Classically, additivity is *derived*: R3 turns disjunction into negated conjunction via De Morgan, and the product rule does the rest. Constructively that route is closed, since $a \vee b \neq \neg(\neg a \wedge \neg b)$ in general:

Theorem 29 (the sum rule is irreducible^{Lean}). *Let Ω be the five-element frame of lower sets of the poset $o < a, o < b$ with a, b incomparable. Then there is a monotone, normalized assignment q on Ω that is additive on disjoint joins yet not modular, and no binary operation S on values satisfies $q(x \sqcup y) = S(q(x), q(y))$ for all x, y . Moreover, the pairs $(\downarrow a, \downarrow a)$ and $(\downarrow a, \downarrow b)$ have identical value-pairs $(\frac{1}{2}, \frac{1}{2})$ but joins valued $\frac{1}{2}$ and 1.*

Unlike for a conjunction, a disjunction's plausibility cannot generally be expressed as its disjuncts' plausibilities via conditioning. So, a constructive Cox theorem *must* posit the sum-rule half. Modularity, which says that the calculus is closed under mixing certainties, is what posits this, justified by Theorem 16. The failure is not repaired by strengthening the logic short of Boolean: the counterexample frame satisfies weak excluded middle^{Lean}, since every nonempty lower set contains the bottom point o , so the irreducibility persists under that intermediate assumption.

6.3 Cox Model Theorem

A regratuation is a map $g : \mathbb{R} \rightarrow [0, \infty]$ with $g(0) = 0$. No such map is strictly monotone on all of $\mathbb{R}$, since strictness would force $g(-1) < g(0) = 0$, impossible in $[0, \infty]$. The repair (strictness only on $[0, 1]$, where plausibilities live) is exactly the same domain restriction that Halpern's critique of the classical theorem turns on. The corrected statement is:

Theorem 30 (the constructive Cox theorem^{Lean}). *For every frame Ω , every Cox model on Ω whose unconditional*

plausibility is modular regratuates to a valuation: there exist $v : \text{Valuation } \Omega$ and g strictly monotone on $[0, 1]$ with $g(0) = 0, g(1) = 1$, and for all $a \in \Omega, v(a) = g(\text{pl}(a, \top))$.

The regratuation for the sum-rule half is straightforward ($g = \text{ENNReal.ofReal}$). The product-rule half, showing F regratuates to multiplication, rests on Aczél's associativity theorem [Aczél 1966], a statement of pure analysis whose type never mentions Ω . `mathlib` has no Aczél theorem, no Hölder-style embedding for ordered semigroups on intervals, and no Cauchy functional equation under monotonicity, so we build the analytic core from scratch, obtaining a normalized, strictly monotone multiplicative generator on the positive cone^{Lean}. The construction also weakens Hölder's classical hypotheses [Hölder 1901]: the Archimedean property is *derived* from continuity and positivity rather than assumed, and the interval form of divisibility follows from the intermediate value theorem^{Lean} applied to a plain divisibility hypothesis^{Lean}. The constructive/classical divide is therefore *entirely* in the sum rule: nothing in the product-rule apparatus mentions Ω . Appendix C gives the construction. The residual analytic gap is discussed in Section 9.

7 Two Grounding Models

Two concrete models demonstrate that our abstract theory is non-degenerate. One shows every classical measure *contains* an intuitionistic probability. The other shows the intuitionistic theory cannot collapse back without solving the halting problem.

7.1 Measures through the Interior Operator

Gödel–McKinsey–Tarski interpret intuitionistic logic in modal S4: propositions are *open* sets, and the modal necessity operator $\Box$ of S4 is interpreted as the topological interior, written $\text{int}(\cdot)$ below [Gödel 1933; McKinsey and Tarski 1944]. Our valuations quantify that same translation:

Theorem 31 (measures as valuations^{Lean}). *Let μ be a probability measure on a space $X, A \subseteq X$, and define $v(A) = \mu(\text{int } A)$. Since $\text{int } U = U$ for open U , v restricts on the frame $\mathcal{O}(X)$ to $U \mapsto \mu(U)$, and this restriction is a valuation.*

Modularity is $\mu(U \cup V) + \mu(U \cap V) = \mu(U) + \mu(V)$. The slack of an open U is then $1 - \mu(U) - \mu(\text{int } U^c)$, which is the measure of the boundary ∂U . We can now prove that, in general, slack is exactly the boundary mass:

Theorem 32 (slack is boundary mass^{Lean}). *For any probability measure μ on a topological space X and any open U , $\text{slack}(U) = \mu(\partial U)$.*

The Heyting complement of U in $\mathcal{O}(X)$ is the interior of its set complement. So, the undecided region, the complement of $U \sqcup \neg U$, is exactly $\overline{U} \setminus U = \partial U$. The general sub-additivity inequality of Section 2 still holds here^{Lean}. This identification of slack with boundary mass also exhibits the modal identification $\text{Bel} = P(\Box \cdot)$ of the DS literature [Ruspini 1986]

arising from the logic itself. Theorem 15 is the converse in the finite case (in the general case, the converse is the open problem, see Section 4).

To characterize weak excluded middle spaces, we give the (standard) definitions required:

Definition 33 (extremally disconnected). A topological space X is *extremally disconnected* if the closure of every open set is again open.

Definition 34 (Stone space, Stonean space). The *Stone space* of a Boolean algebra B is the space of ultrafilters on B (equivalently, the 2-valued homomorphisms $B \rightarrow \{0, 1\}$), topologized by the clopen basis $\{U : b \in U\}$ for $b \in B$. *Stone duality* [Johnstone 1982] identifies Boolean algebras with exactly the compact, Hausdorff, totally disconnected spaces arising this way. A *Stonean space* is a compact Hausdorff space that is also extremally disconnected. Equivalently, it is the Stone space of a *complete* Boolean algebra.

Remark 35 (weak excluded middle on opens^{Lean}). Section 2 noted that imposing weak excluded middle on the whole frame collapses slack to pure double-negation instability. For $\Omega = O(X)$, this global condition holds exactly when X is extremally disconnected.

An immediate corollary gives the characterization:

Corollary 36 (Stonean characterization^{Lean}). *Given a compact Hausdorff space X , weak excluded middle on $O(X)$ holds exactly when X is a Stonean space.*

7.2 Slack Does Not Determine the Valuation

Submodularity (Theorem 7) is a necessary condition on slack. It is not sufficient: distinct valuations, each satisfying every axiom a valuation must satisfy, can share a slack function.

Example 37 (boundaries in the three-point space^{Lean}). Let $\text{ThreePt} = \{p_A, p_B, p_Z\}$ carry the topology generated by the two singletons $\{p_A\}$ and $\{p_B\}$, so its opens are exactly $\emptyset$, $\{p_A\}$, $\{p_B\}$, $\{p_A, p_B\}$, and the whole space. For every open U , $\partial U \subseteq \{p_Z\}$.

This is a fact about the topology alone, independent of any measure: p_Z has no proper open neighbourhood, so it is the only point any open's frontier can contain.

Theorem 38 (non-recovery^{Lean}). *There exist valuations $v \neq v'$ on $O(\text{ThreePt})$ with $\text{slack}(v) = \text{slack}(v')$ as functions.*

For weights $a, b, c \geq 0$ with $a + b + c = 1$, the point measure $\mu_{a,b,c} = a \delta_{p_A} + b \delta_{p_B} + c \delta_{p_Z}$, read through Section 7.1's bridge, gives a valuation $v_{a,b,c}$ on $O(\text{ThreePt})$. By Example 37 and Theorem 32, its slack at an open U is $\mu_{a,b,c}(\partial U)$, equal to c when $p_Z \in \partial U$ and to 0 otherwise: a quantity depending only on c , not on how the remaining mass splits between a and b . So $v_{1/4,1/4,1/2}$ and $v_{1/8,3/8,1/2}$ have identical slack functions while disagreeing at $\{p_A\}$, where they take the values $1/4$

and $1/8$. Both are fully modular, monotone, and normalized: nothing in Theorem 7 or its proof distinguishes them.

Weak excluded middle does not rescue recoverability. If anything, the fully classical case is the worst one for it. When X is extremally disconnected, the De Morgan gap vanishes (Section 7.1) and slack degenerates to the double-negation gap alone, but on a space where excluded middle holds outright that gap vanishes too, so slack is identically 0 for every valuation. The zero slack function is then shared by every probability measure on X , which is the most extreme failure of recovery available.

7.3 The Computability Guard

The theory developed here is non-degenerate only if there are spaces whose slack does not vanish identically. Computability supplies them:

Definition 39 (Sierpiński topology, semidecidable proposition). The *Sierpiński space* $\mathbb{S}$ is the type Prop of propositions carrying the topology with exactly three opens: $\emptyset$, the set $\text{halts} = \{q \mid q\}$ of true propositions, and $\mathbb{S}$ itself [Escardó 2004; Rosolini 1986]. A proposition is *semidecidable* if it is an open in this topology.

The topology encodes verifiability. A point of $\mathbb{S}$ is a proposition, say “this computation halts.” The one nontrivial observation available is membership in halts , established by producing a proof (running the computation to termination). No open captures the complementary observation, and on the frame $O(\mathbb{S}) = \{\perp, \text{halts}, \top\}$ that Σ_1 asymmetry is a theorem:

Theorem 40 (the Σ_1 asymmetry^{Lean}). $\neg \text{halts} = \perp$.

Every nonempty open of $\mathbb{S}$ contains True, so $\perp$ is the only open disjoint from halts . Thus, no nonempty open of $\mathbb{S}$ is evidence that the computation does not halt.

We now instantiate the measure-to-valuation construction of Theorem 31 at $X = \mathbb{S}$, allowing us to compute the slack:

Example 41 (halting slack^{Lean}). Let $\mu = p \cdot \delta_{\text{True}} + (1-p) \cdot \delta_{\text{False}}$, and let v be the resulting valuation. Then

$$\text{slack}(\text{halts}) = 1 - p = \mu(\partial\{q \mid q\}) = \mu\{\text{False}\}.$$

Since halts is already open, $v(\text{halts}) = \mu(\text{halts}) = p$. The DS “ignorance” of Section 5 is here the mass of the single point False, the one outcome no finite observation can ever resolve. This probability is a fact about the model, not about anyone's ignorance.

Chaitin's Ω . The same bridge scales to algorithmic information theory. Fix a universal prefix-free machine U , *prefix-free* in that no halting program of U is a proper prefix of another, and *universal* in that it simulates every other prefix-free machine. Give 2^ω , the space of infinite bit strings, the fair-coin measure μ . The event H of having a halting program of U as a prefix is open, since halting is established by

a finite prefix (Definition 39), and prefix-freeness makes H a disjoint union of cylinders, so $v(H) = \mu(H) = \sum_{p \text{ halts}} 2^{-|p|}$; exactly Chaitin’s halting probability Ω [Chaitin 1975]. Its slack $\mu(\partial H)$ is the mass of strings approximated arbitrarily well by halting strings while never having a halting program as a prefix, and Ω ’s classical uncomputability is the aggregate form of the pointwise obstruction above. The measure-theoretic core is mechanized on Cantor space with mathlib’s infinite product measure: basic cylinders are clopen with measure $2^{-|w|}^{\text{Lean}}$, and a *finite* prefix-free set of words has pairwise disjoint cylinders of total measure $\sum_w 2^{-|w|}^{\text{Lean}}$, the Kraft-inequality content underlying Ω . The universal machine itself, and the countable extension of the finite Kraft sum, are computability theory beyond our scope.

Definition 42 (halting problem). For a code c and input n , the *halting problem* is the predicate $H(c, n) = \text{“the machine with code } c \text{ halts on input } n\text{”}$ (ComputablePred).

Slack cannot be eliminated by simply declaring a classical answer for each code. The family of valuations defined below attempts to do this:

Definition 43 (sharp classical readout). For a given n , the *sharp classical readout* is the family of slack-free valuations $\rho_n(c) \in \text{Valuation}(\mathcal{O}(\mathbb{S}))$, with $\rho_n(c)(\text{halts}) = 1$ if $H(c, n)$ holds, and $\rho_n(c)(\text{halts}) = 0$ otherwise.

However, this family is provably not computable:

Theorem 44 (the sharp readout is uncomputable^{Lean}). For every n , the predicate $c \mapsto (\rho_n(c)(\text{halts}) = 1)$ is not computable.

The proof identifies the predicate with $H(\cdot, n)$ and applies the undecidability of the halting problem (ComputablePred.halting_problem).

We can conclude that slack is not a defect of the constructive theory: any axiom strong enough to force classical collapse demands that its models compute the uncomputable. For the sum rule, this guard plays the role the explicit Cox-model instance plays for the product rule (Section 6): each axiom set in the paper comes with non-trivial models satisfying it.

8 Related Work

Informal precedents. Pairing probability with intuitionistic logic is not a new idea, though the literature on it is scattered. Weatherson develops intuitionistic probability axiomatically and defends it with Dutch-book arguments [Weatherson 2003]. His axioms match our valuation axioms exactly, with negation derived rather than assumed, so the object itself and the bound $v(a) + v(\neg a) \leq 1$ are his. What is new here is the decomposition of that bound’s slack into dnGap and deMorganGap and its determinacy from v alone (Theorem 6). Paris gives an informal theorem that any modular assignment of non-classical truth values is exactly a Dempster–Shafer belief function [Paris 1994]. We

mechanize the analogous fact for our intuitionistic setting: the 2-monotone case follows directly from modularity^{Lean}, and the full tower is Theorem 23. The identification $\text{Bel} = P(\Box \cdot)$ connecting belief functions [Dempster 1967; Shafer 1976] to modal logic is established [Ruspini 1986]. Section 7 shows this identification is not a separate axiom for us but a special case of our frame-level slack, recovered automatically once the frame is a topology and $\neg$ is read as the interior of the complement^{Lean}. On the localic side, valuations on frames are standard objects of constructive measure theory, with Vickers’s work and the Coquand–Spitters development as canonical references [Coquand and Spitters 2009; Vickers 2008]. These normally build in Scott continuity and target additive, Riesz-style goals. Section 4 shows Scott continuity to be a substantive assumption (it decides point-representability on the frames we examine), so our base structure omits it. Cox, Jaynes, Halpern’s counterexample, and the Van Horn and Arnborg–Sjödín repairs [Arnborg and Sjödín 2000; Cox 1946; Halpern 1999; Horn 2003; Jaynes 2003] are classical throughout, and Cox’s and Van Horn’s derivations both lean on double-negation elimination. Section 6 proves the sum-rule half without that move, at the price of positing modularity as an axiom rather than deriving it (Theorem 29). Van Horn observes that dropping axiom R3 yields an interval-valued theory, exactly the belief/plausibility pair our slack separates. To our knowledge, no prior work, informal or formal, states either characterization theorem: complement rule $\iff$ excluded middle (Theorem 10), or Dempster = Bayes $\iff$ zero slack at the evidence (Theorem 27).

Prior formalizations. Two bodies of mechanized work are adjacent, and neither overlaps. ALEA [Audebaud and Paulin-Mohring 2009] formalizes a discrete distribution monad in Coq for randomized programs: additive probability, no event logic in our sense. Closest is the HoTT/Coq line of Faissole and Spitters on *valuations* over lower reals [Faissole and Spitters 2017], done constructively with Scott continuity and aimed at integration (Riesz) and a Giry-style valuation *monad*. Their valuations differ from ours in carrying Scott continuity, and their monad is the Scott-continuous instance of the question examined in Section 9. Our development drops the continuity their base structure builds in, and the two mechanizations do not otherwise overlap.

Non-Boolean probability elsewhere. Quantum probability on orthomodular lattices, with Gleason’s theorem [Birkhoff and von Neumann 1936; Gleason 1957], is the long-standing precedent for re-basing probability on a non-Boolean event logic. Orthomodular lattices give up distributivity itself, while our frames stay distributive by definition (Definition 1), locating all the non-classical behavior in the failure of excluded middle alone. Free probability, fuzzy measures, and imprecise probability [Walley 1991]

vary other parameters (independence, additivity, single-valuedness). Where Walley's coherent lower previsions posit the belief/plausibility gap directly as a coherence primitive, our slack (Section 2) is derived, forced by modularity and negation rather than assumed. Fritz's synthetic Markov-category framework [Fritz 2020] varies the ambient structure categorically rather than order-theoretically, axiomatizing conditional independence and sufficient statistics abstractly across discrete, measure-theoretic, and Gaussian probability alike. It remains unmechanized and is orthogonal to our variation in the underlying logic. Wolpert and Kinney model mathematics itself as a stochastic process, replacing binary theoremhood with a probability distribution [Wolpert and Kinney 2024]. Theoremhood is a semidecidable event, so the computability guard of Section 7 bears directly on that program: slack-free credences about such events are uncomputable. Our variation is specifically the *logic*, holding the Cox-style methodology fixed. Within DS theory, the two conditionings and their inequivalence are classical [Fagin and Halpern 1991; Shafer 1976]. What is new in Theorem 27 is the exact logical characterization of their agreement, and its mechanization.

9 Conclusion

Many properties of programs are only semidecidable, so the Boolean event algebra Kolmogorov's axioms [Kolmogorov 1933] presuppose is not a good fit for reasoning about them. The notion of graded belief does not presuppose Boolean logic. Classicality is imposed only via the complement rule (Theorem 10). Its dynamic counterpart is the agreement of Dempster and Bayesian conditioning (Theorem 27). Everything else survives the intuitionistic substrate intact: conditioning, Bayes symmetry, disjoint additivity, total probability over genuine partitions, and inclusion-exclusion itself (Theorem 22). The weaker logic adds the slack structure, along with its two-gap decomposition, and the atomic/diffuse divide. For this new structure, we demonstrate a belief-function reading through $\neg\neg$, and give a computability-theoretic reason that structure cannot be dispensed with: forcing it away on a semidecidable event is exactly as hard as solving the halting problem (Section 7).

Functoriality and the monad question. For a fixed choice of logic, the connection to the categorical picture is concrete. Pushforward along a frame homomorphism $f : \Omega \rightarrow \Omega'$ sends a valuation v on Ω' to $(f_*v)(a) = v(f(a))$ on Ω , and this preserves all four valuation axioms, making $\text{Val}(-)$ a functor on locales, the localic counterpart of the functor part of the Giry monad [Giry 1982]. This is mechanized end to end, the pushforward together with the two functor laws checked against identity and composition of frame homomorphisms^{Lean}. The point valuations δ_p of Section 4 are the unit, already formalized, and Theorem 15 shows

that on finite frames Val agrees, on objects, with the classical finite-distribution monad on points. The multiplication is where the substance is. Flattening a valuation of valuations is integration, hence product valuations and Fubini, and by Kock's identification of commutative monads with Fubini theorems [Kock 2012], whether statistics-as-kernel-composition transports to a weaker logic is precisely the question whether that logic's valuation functor is a commutative monad. Two machine-checked results delimit the multiplication from both sides.

On the chain product, the question of *which* product valuation to take does not arise, since the frame $\text{Low}(\mathbb{N} \times \mathbb{N})$ is *rigid*: every lower set is a *finite* union of rectangles, because beyond a stabilization height every occupied column is full, so the full columns and finitely many rows account for everything^{Lean}. The rigidity mechanism is general. By the inclusion-exclusion equality, two valuations agreeing on any $\sqcap$ -closed family agree on all finite joins of its members^{Lean}, and the chain product is the instance where that family is the rectangles. The product valuation there, whenever it exists, is therefore unique, with no continuity assumed^{Lean}.

On the powerset frame $\text{Set}(\mathbb{N} \times \mathbb{N})$, whose elements are far from finite unions of rectangles, non-uniqueness is a theorem. The indicator of an ultrafilter is a valuation^{Lean}, and any ultrafilter on a product splits rectangles through its marginal pushforwards^{Lean}. Taking a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the two iteration orders of its tensor with itself and its diagonal pushforward carry identical values on *every* rectangle, yet they differ on the upper triangle $\{(m, m') \mid m < m'\}$ ^{Lean}. The two orders of iterated integration of the same marginal data therefore disagree: Fubini fails, concretely^{Lean}. Compare the Fubini pathologies of non-additive integration [Denneberg 1994; Ghirardato 1997]. By Kock's correspondence, no canonical commutative multiplication exists for unrestricted valuations. On the powerset frame these valuations are finitely additive probability charges in the sense of [Rao and Rao 1983], so this half of the dichotomy is not specific to intuitionism: the freedom is already present in the classical finitely additive setting, and dropping continuity is what admits it. A monad therefore requires restricting either the frames or the valuations.

The obstruction to a monad. The question upstream of the multiplication is worth stating precisely, though what follows is an assessment rather than a theorem. A monad needs a category for Val to be an endofunctor on. The space of valuations must itself be an object, a "space" whose opens one can speak of. For Scott-continuous valuations that space is a locale and monad structure is available, constructively, through the probabilistic powerdomain line [Coquand and Spitters 2009; Jones and Plotkin 1989; Vickers 2008]. Even there, exhibiting a cartesian closed category closed under the construction is the Jung-Tix problem, open since 1998 for the classical probabilistic powerdomain functor itself [Jung

and Tix 1998], though reframed versions of the category have recently been settled. Dropping continuity makes the difficulty prior rather than merely worse, since the standard presentations of the valuation space use continuity essentially.

One candidate route to a well-posed question does exist. The localic treatments present the space of continuous valuations as the classifying locale of a propositional geometric theory [Coquand and Spitters 2009; Vickers 2008], with generators asserting $q < v(a)$ for rational q and axioms for monotonicity, normalization, modularity (expressible with rational parameters and countable disjunctions), and continuity. Deleting the continuity axiom leaves a geometric theory, so a classifying locale still exists on general grounds, and classically its points are exactly the valuations of this paper. We have verified none of the structure this candidate would need. The unit looks unproblematic, but the multiplication requires integrating the evaluation maps against a merely modular, discontinuous valuation, where the sum of two lower-real-valued maps is an infinite join, and the diffuse part of Example 17 is precisely what such joins fail to capture. We offer the formulation as a target, not a result.

A more modest strategy, with a better track record, is to carve out a fragment rather than construct the whole monad. On the measure-theoretic side, the failure of cartesian closure for measurable spaces was answered by quasi-Borel spaces [Heunen et al. 2017]. On the domain-theoretic side, commutativity of the full valuations monad on DCPO is open, and the response is a hierarchy of commutative *submonads* generated from simple valuations [Jia et al. 2021]. Our decomposition theorem names the corresponding fragment here: the atomic part. As a first machine-checked step in that direction, we show that the countably-presented fragment composes correctly with no space structure needed at all. A countable mixture of valuations is a valuation^{Lean}, a mixture of mixtures flattens to a mixture with product weights, which is discrete Chapman–Kolmogorov^{Lean}, the one-point mixture is the identity^{Lean}, and iterated countable mixing is order-independent^{Lean}, all by the unconditional Fubini of $[0, \infty]$ sums over sigma types. This is associative, unital, commutative kernel composition for countably-atomic data, in direct contrast with the non-uniqueness above. It is not yet a monad: the presentations are not canonical, we do not quotient by “presents the same valuation,” and the diffuse part is exactly what no countable presentation captures.

In summary, the unit, the finite level, and countable kernel composition with its commutativity are proved. Rigidity on chain products and non-uniqueness on the powerset frame are proved, so the general multiplication is settled: it is not canonically determined. What remains is the ambient category, a formulation problem before it is anything else.

Open problems and future work. Four problems are open: (1) representing the diffuse part on non-spatial frames,

(2) completing the Aczél development (continuity of the generator and its extension beyond the cone), (3) deriving modularity from a criterion more primitive than the mixture characterization (Theorem 16), knowing it cannot come from disjunction data alone (Theorem 29), and (4) a machine-checked Dutch-book theorem in the style of [Weatherson 2003], which to our knowledge no proof assistant has for *any* logic. On the monad, two continuations are concrete: quotient the countable presentations into a Kleisli category of countably-atomic kernels, and mechanize the classifying-locale formulation above, which needs classifying locales of geometric theories, infrastructure mathlib currently lacks. Both are confined to a fragment, since the unrestricted multiplication is provably not canonical. Beyond intuitionistic logic there is no ambient category in which “logic $\mapsto$ its probability theory” is a functor. What transfers is the method: state the classical desiderata, delete the axioms the target logic no longer licenses, and let a proof assistant check what survives, what splits, and what becomes unsatisfiable. Linear logic is the natural next case: its event algebras are quantales, a valuation there should grade the consumption of evidence, and the analogues of our representation theorems and of slack, for the failure of contraction, are open [Danos and Ehrhard 2011].

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A Values in the Extended Reals

Valuations take values in the extended non-negative reals $[0, \infty]$, following the measure-theoretic convention of mathlib. Two consequences of this choice are used silently in the main text. First, sums indexed by an arbitrary family are unconditionally defined, with no summability hypothesis, and they commute over dependent sums: for a function f on a sigma type $\Sigma a, \beta(a)$, one has $\sum_p f(p) = \sum_a \sum_b f(a, b)$ with no side condition. This is what lets the countable-mixture results of Section 9 hold as stated. Second, subtraction is truncated at 0. This never affects the main text, since every subtraction there is applied to a pair already ordered by monotonicity, as with the mass of Section 4, so the truncation is never triggered.

B Lean Statements

B.1 Valuations on Frames

```
structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b =
    toFun (a ⊔ b) + toFun (a ⊓ b)

theorem Valuation.add_compl_le_one (v :
  Valuation Ω) (a : Ω) :
  v a + v aᶜ ≤ 1

theorem Valuation.add_compl_eq_sup (v :
  Valuation Ω) (a : Ω) :
  v a + v aᶜ = v (a ⊔ aᶜ)

theorem Valuation.le_one (v : Valuation Ω)
  (a : Ω) : v a ≤ 1

theorem Valuation.
  slack_eq_dnGap_add_deMorganGap (v :
  Valuation Ω) (a : Ω) :
  v.slack a = v.dnGap a + v.deMorganGap
  a
```

```

1431 theorem Valuation.slack_eq_sub_add_sub (v
1432   : Valuation  $\Omega$ ) (a :  $\Omega$ ) :
1433   v.slack a = (v acc - v a) + (1 - (v a
1434     cc + v ac))
1435
1436 theorem Valuation.le_compl_compl (v :
1437   Valuation  $\Omega$ ) (a :  $\Omega$ ) : v a ≤ v acc
1438
1439 theorem Valuation.slack_submodular (v :
1440   Valuation  $\Omega$ ) (a b :  $\Omega$ ) :
1441   v.slack (a  $\sqcup$  b) + v.slack (a  $\sqcap$  b) ≤ v.
1442     slack a + v.slack b
1443
1444 def Order.Frame.IsDeMorgan ( $\Omega$  : Type*) [
1445   Order.Frame  $\Omega$ ] : Prop :=
1446    $\forall$  a :  $\Omega$ , acc  $\sqcup$  ac =  $\top$ 
1447
1448 theorem Valuation.
1449   slack_eq_dnGap_of_isDeMorgan
1450   (h $\Omega$  : Order.Frame.IsDeMorgan  $\Omega$ ) (v :
1451     Valuation  $\Omega$ ) (a :  $\Omega$ ) :
1452     v.slack a = v.dnGap a
1453
1454 noncomputable def Valuation.condVal (v :
1455   Valuation  $\Omega$ ) (b :  $\Omega$ ) (hb : v b ≠ 0) :
1456   Valuation  $\Omega$ 
1457
1458 theorem Valuation.condVal_mul (v :
1459   Valuation  $\Omega$ ) (b :  $\Omega$ ) (hb : v b ≠ 0) (a
1460   :  $\Omega$ ) :
1461   v.condVal b hb a * v b = v (a  $\sqcap$  b)
1462
1463 theorem Valuation.condVal_symm (v :
1464   Valuation  $\Omega$ ) {a b :  $\Omega$ }
1465   (ha : v a ≠ 0) (hb : v b ≠ 0) :
1466   v.condVal b hb a * v b = v.condVal a
1467     ha b * v a
1468
1469 theorem Valuation.cond_add_compl_le (v :
1470   Valuation  $\Omega$ ) (a b :  $\Omega$ ) :
1471   v (a  $\sqcap$  b) + v (ac  $\sqcap$  b) ≤ v b
1472
1473 theorem Valuation.
1474   add_compl_eq_one_of_complemented (v :
1475     Valuation  $\Omega$ ) {a :  $\Omega$ }
1476     (ha : a  $\sqcup$  ac =  $\top$ ) : v a + v ac = 1
1477
1478 variable [CompleteBooleanAlgebra  $\Omega$ ]
1479
1480 theorem classical_additivity (v :
1481   Valuation  $\Omega$ ) (a :  $\Omega$ ) :
1482   v a + v ac = 1
1483
1484 variable [CompleteBooleanAlgebra  $\Omega$ ]
1485

```

```

theorem classical_slack_zero (v :
Valuation  $\Omega$ ) (a :  $\Omega$ ) :
v.slack a = 0

noncomputable def Valuation.mix (w :  $\iota \rightarrow \mathbb{R}$ 
 $\geq 0$ ) (hw :  $\sum i, w i = 1$ )
(v :  $\iota \rightarrow$  Valuation  $\Omega$ ) : Valuation  $\Omega$ 

```

B.2 Where Classicality Lives

```

def Valuation.HasClassicalNegation (v :
Valuation  $\Omega$ ) : Prop :=
 $\forall$  a, v a + v ac = 1

theorem hasClassicalNegation_of_em (hem :
 $\forall$  a :  $\Omega$ , a  $\sqcup$  ac =  $\top$ )
(v : Valuation  $\Omega$ ) : v.
HasClassicalNegation

theorem em_of_forall_hasClassicalNegation
(h :  $\forall$  v : Valuation  $\Omega$ , v.
HasClassicalNegation) :  $\forall$  a :  $\Omega$ , a
 $\sqcup$  ac =  $\top$ 

theorem sharp_iff_point (v : Valuation  $\Omega$ )
:
v.IsSharp  $\leftrightarrow \exists$  (J : Order.Ideal  $\Omega$ ) (hJ
: J.IsPrime) (htop :  $\top \notin J$ ),
v = Ideal.toValuation J hJ htop

theorem
sharp_scott_iff_completelyPrimePoint (v
: Valuation  $\Omega$ ) :
v.IsSharp  $\wedge$  v.ScottContinuous  $\leftrightarrow$ 
 $\exists$  (J : Order.Ideal  $\Omega$ ) (hJ : Ideal.
IsCompletelyPrime J) (htop :  $\top \notin$ 
J),
v = Ideal.toValuation J hJ.1 htop

```

B.3 Valuation Representation

```

noncomputable def Valuation.mass (v :
Valuation (LowerSet P)) (p : P) :  $\mathbb{R}_{\geq 0}$ 

theorem Valuation.eq_sum_mass (v :
Valuation (LowerSet P)) (U : LowerSet P
) :
v U =  $\sum p \in U.toFinset, v.mass p$ 

theorem Valuation.sum_mass (v : Valuation
(LowerSet P)) :  $\sum p, v.mass p = 1$ 

```

1541	noncomputable def Valuation.toPMF (v :	(a : ι → Ω) : v (s.sup a) + v.iesEven	1596
1542	Valuation (LowerSet P)) : PMF P	s a = v.iesOdd s a	1597
1543			1598
1544	theorem Valuation.exists_pmf_presentation	theorem Valuation.infty_monotone (v :	1599
1545	{Θ : Type u} [Order.Frame Θ] [Finite Θ]	Valuation Ω) (s : Finset ι) (a : ι → Ω)	1600
1546	(v : Valuation Θ) :	:	1601
1547	∃ (P : Type u) (⊥ : PartialOrder P) (⊥	v.iesOdd s a ≤ v ((s.sup a) ^{cc}) + v.	1602
1548	: Fintype P) (⊥ : DecidableEq P)	iesEven s a	1603
1549	(e : Θ ≈o LowerSet P),		1604
1550	(∀ a, v a = ∑ p ∈ (e a).toFinset, (v	theorem Valuation.two_monotone (v :	1605
1551	.mapEquiv e).mass p) ∧	Valuation Ω) (a b : Ω) :	1606
1552	∑ p, (v.mapEquiv e).mass p = 1	v a + v b ≤ v (a ⊔ b) ^{cc} + v (a ⊓ b)	1607
1553			1608
1554	noncomputable def deltaPoint (p : P) :	theorem Valuation.plausibility_sub_self (v	1609
1555	Valuation (LowerSet P)	: Valuation Ω) (a : Ω) :	1610
1556		v.plausibility a - v a = v.slack a	1611
1557	theorem Valuation.eq_mix_deltaPoint (v :		1612
1558	Valuation (LowerSet P)) :	theorem Valuation.sup_compl_decomp (v :	1613
1559	v = Valuation.mix v.mass v.sum_mass (	Valuation Ω) (a b : Ω) :	1614
1560	fun p => deltaPoint p)	v (a ⊔ b ^c) = v b ^c + (v (a ⊓ b) + v.	1615
1561		emGap a b)	1616
1562	theorem		1617
1563	exists_valuation_not_point_representable	noncomputable def Valuation.dempsterCond (	1618
1564	:	v : Valuation Ω) (b : Ω)	1619
1565	∃ v : Valuation (LowerSet ℕ), ∑' p, v.	(hb : v b ≠ 0) : Valuation Ω	1620
1566	mass p ≠ v ⊤		1621
1567		theorem Valuation.dempsterCond_apply (v :	1622
1568	theorem topIndicator_mass (n : ℕ) :	Valuation Ω) (b : Ω) (hb : v b ≠ 0)	1623
1569	topIndicator.mass n = 0	(a : Ω) : v.dempsterCond b hb a = (v (	1624
1570		a ⊔ b ^c) - v b ^c) / (1 - v b ^c)	1625
1571	theorem topIndicator_not_scott :	theorem Valuation.	1626
1572	(⊔ n, topIndicator (LowerSet.Iic n)) ≠	dempsterCond_eq_condVal_iff_slack (v :	1627
1573	topIndicator ⊤	Valuation Ω) {b : Ω}	1628
1574		(hb : v b ≠ 0) :	1629
1575	theorem eq_tsum_mass_of_scott (v :	(∀ a, v.dempsterCond b hb a = v.	1630
1576	Valuation (LowerSet ℕ))	condVal b hb a) ↔ v.slack b = 0	1631
1577	(hsc : v ⊤ = ⊔ n, v (LowerSet.Iic n))		1632
1578	: v ⊤ = ∑' n, v.mass n		1633
1579			1634
1580	variable [∀ p : P, Finite (Set.Iic p)]	B.5 Cox's Theorem	1635
1581			1636
1582	theorem Valuation.isPurelyAtomic_of_scott	structure CoxModel (Ω : Type*) [Order.	1637
1583	(v : Valuation (LowerSet P))	Frame Ω] where	1638
1584	(hsc : v ⊤ = ⊔ (S : Finset P), v (S.	pl : Ω → Ω → ℝ	1639
1585	sup LowerSet.Iic)) : IsPurelyAtomic	mono_left : ∀ c, Monotone fun a => pl a	1640
1586	v	c	1641
1587		pl_bot : ∀ c, pl ⊥ c = 0	1642
1588	theorem tsum_mass_le (v : Valuation (	pl_top : ∀ c, pl ⊤ c = 1	1643
1589	LowerSet P)) : ∑' p, v.mass p ≤ v ⊤	F : ℝ → ℝ → ℝ	1644
1590		conj : ∀ a b c, pl (a ⊓ b) c = F (pl a (	1645
1591		b ⊓ c)) (pl b c)	1646
1592	B.4 The Dempster-Shafer Reading	F_assoc : ∀ x y z, F (F x y) z = F x (F	1647
1593		y z)	1648
1594	theorem Valuation.inclusion_exclusion (v :	F_strictMono_left : ∀ y, 0 < y →	1649
1595	Valuation Ω) (s : Finset ι)	StrictMono fun x => F x y	1650

```

1651 theorem isDeMorgan_lowerSetV : Order.Frame
1652   .IsDeMorgan (LowerSet V)
1653
1654 theorem modularity_irreducible :
1655    $\exists (\Omega : \text{Type}) (\_ : \text{Order.Frame } \Omega) (q : \Omega \rightarrow \mathbb{R}_{\geq 0\infty}),$ 
1656   Monotone q  $\wedge$  q  $\perp = 0 \wedge q \top = 1 \wedge$ 
1657    $(\forall x y : \Omega, x \sqcap y = \perp \rightarrow q(x \sqcup y) = q x + q y) \wedge$ 
1658    $\neg (\forall x y : \Omega, q x + q y = q(x \sqcup y) + q(x \sqcap y))$ 
1659
1660 theorem no_disjunction_functional :
1661    $\neg \exists S : \mathbb{R}_{\geq 0\infty} \rightarrow \mathbb{R}_{\geq 0\infty} \rightarrow \mathbb{R}_{\geq 0\infty},$ 
1662    $\forall x y : \text{LowerSet } V, qV(x \sqcup y) = S(qV x)(qV y)$ 
1663
1664 theorem constructive_cox (M : CoxModel  $\Omega$ )
1665   (hmod :  $\forall a b : \Omega, M.\text{pl } a \top + M.\text{pl } b \top = M.\text{pl } (a \sqcup b) \top + M.\text{pl } (a \sqcap b) \top$ )
1666   :
1667    $\exists (v : \text{Valuation } \Omega) (g : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0\infty}),$ 
1668   StrictMonoOn g (Set.Icc 0 1)  $\wedge$  g 0 = 0  $\wedge$  g 1 = 1  $\wedge$ 
1669    $\forall a, v a = g(M.\text{pl } a \top)$ 
1670
1671 noncomputable def coxModelENNReal :
1672   CoxModel  $\mathbb{R}_{\geq 0\infty}$ 
1673
1674 theorem Scale.exists_mul_generator :
1675    $\exists G : \mathbb{R} \rightarrow \mathbb{R},$ 
1676    $(\forall x y, S.u \leq x \rightarrow S.u \leq y \rightarrow G(S.F x y) = G x * G y) \wedge$ 
1677    $(\forall x y, S.u \leq x \rightarrow x < y \rightarrow G x < G y)$ 
1678
1679 theorem Scale.aczelStatement_cone :
1680    $\exists G : \mathbb{R} \rightarrow \mathbb{R},$ 
1681    $(\forall x y, S.u \leq x \rightarrow S.u \leq y \rightarrow G(S.F x y) = G x * G y) \wedge$ 
1682   StrictMonoOn G (Set.Ici S.u)
1683
1684 theorem exists_diag_eq {F :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ } {a b c :  $\mathbb{R}$ } (hab : a  $\leq$  b)
1685   (hcont : ContinuousOn (fun x => F x x) (Set.Icc a b))
1686   (hc : c  $\in$  Set.Icc (F a a) (F b b)) :  $\exists x \in \text{Set.Icc } a b, F x x = c$ 
1687
1688 structure Scale where
1689   F :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 
1690   hdiv :  $\forall c, \exists x, F x x = c$ 
1691   hassoc :  $\forall x y z, F(F x y) z = F x (F y z)$ 
1692   hpos :  $\forall c x, x < F x c$ 

```

```

hmono :  $\forall x_1 x_2 y_1 y_2 : \mathbb{R}, x_1 \leq x_2 \rightarrow y_1 \leq y_2 \rightarrow F x_1 y_1 \leq F x_2 y_2$ 
hcont : Continuous fun p :  $\mathbb{R} \times \mathbb{R} \Rightarrow F p$  .1 p.2
hbot :  $\forall x, \text{Filter.Tendsto } (\text{fun } z \Rightarrow F x z) \text{Filter.atBot } (\text{nhds } x)$ 
u :  $\mathbb{R}$ 

```

B.6 Two Grounding Models

```

noncomputable def MeasureTheory.Measure.toValuationOpens
  ( $\mu : \text{Measure } X$ ) [IsProbabilityMeasure  $\mu$ ] : Valuation (Opens X)
theorem toValuationOpens_slack_eq_frontier
  ( $\mu : \text{Measure } X$ ) [IsProbabilityMeasure  $\mu$ ] (U : Opens X) :
   $\mu.\text{toValuationOpens.slack } U = \mu(\text{frontier } (U : \text{Set } X))$ 
theorem interiorMeasure_add_compl_le ( $\mu : \text{Measure } X$ ) [IsProbabilityMeasure  $\mu$ ]
  (S : Set X) :  $\mu.\text{interiorMeasure } S + \mu.\text{interiorMeasure } S^c \leq 1$ 
theorem
  isDeMorgan_opens_iff_extremallyDisconnected :
  Order.Frame.IsDeMorgan (Opens X)  $\leftrightarrow$  ExtremallyDisconnected X
theorem isDeMorgan_opens_iff_stonean [CompactSpace X] [T2Space X] :
  Order.Frame.IsDeMorgan (Opens X)  $\leftrightarrow$  ExtremallyDisconnected X
inductive ThreePt : Type
  | ptA : ThreePt
  | ptB : ThreePt
  | ptZ : ThreePt
theorem frontier_subset_ptZ {U : Set ThreePt} (hU : IsOpen U) :
  frontier U  $\subseteq$  {ptZ}
theorem exists_ne_valuation_same_slack :
   $\exists v v' : \text{Valuation } (\text{Opens ThreePt}), v.\text{slack} = v'.\text{slack} \wedge v \neq v'$ 
theorem haltsOpen_compl : haltsOpenc = ( $\perp : \text{Opens Prop}$ )

```

```

1761 theorem slack_eq_boundary {p :  $\mathbb{R}_{\geq 0\infty}$ } (hp1
1762   : p ≤ 1) :
1763   (sierpinskiValuation hp1).slack
1764     haltsOpen
1765     = coinMeasure p (frontier {q : Prop
1766       | q})
1767
1768 theorem sharpReadout_not_computable (n :  $\mathbb{N}$ 
1769   ) :
1770   ¬ComputablePred fun c : Code =>
1771     sharpReadout n c haltsOpen = 1
1772
1773 theorem isOpen_wordCylinder (w : List Bool
1774   ) : IsOpen (wordCylinder w)
1775
1776 theorem cantorMeasure_wordCylinder (w :
1777   List Bool) :
1778   cantorMeasure (wordCylinder w) = (1 /
1779     2) ^ w.length
1780
1781 theorem
1782   disjoint_wordCylinder_of_not_prefix {v
1783     w : List Bool}
1784   (hv : ¬ v <+: w) (hw : ¬ w <+: v) :
1785   Disjoint (wordCylinder v) (
1786     wordCylinder w)
1787
1788 theorem cantorMeasure_biUnion_wordCylinder
1789   {F : Finset (List Bool)}
1790   (hF : PrefixFree F) :
1791   cantorMeasure ( $\bigcup w \in F$ , wordCylinder w
1792     ) =  $\sum w \in F$ , (1 / 2 :  $\mathbb{R}_{\geq 0\infty}$ ) ^ w.
1793     length
1794
1795
1796 B.7 The Monad Question
1797
1798 def Valuation.comap (f : FrameHom  $\Gamma$   $\Omega$ ) (v
1799   : Valuation  $\Omega$ ) : Valuation  $\Gamma$ 
1800
1801 theorem Valuation.comap_id (v : Valuation
1802    $\Omega$ ) : v.comap (FrameHom.id  $\Omega$ ) = v
1803
1804 theorem Valuation.comap_comp (f : FrameHom
1805    $\Gamma$   $\Omega$ ) (g : FrameHom  $\Delta$   $\Gamma$ )
1806   (v : Valuation  $\Omega$ ) : v.comap (f.comp g)
1807     = (v.comap f).comap g
1808
1809 theorem exists_rect_decomposition (W :
1810   LowerSet ( $\mathbb{N} \times \mathbb{N}$ )) :
1811    $\exists$  (N :  $\mathbb{N}$ ) (A B :  $\mathbb{N} \rightarrow$  LowerSet  $\mathbb{N}$ ),
1812     W = (Finset.range N).sup fun i => A
1813       i  $\times^s$  B i
1814
1815

```

```

theorem Valuation.eq_on_finsetSup_of_eq_on
  {G : Set  $\Omega$ }
  (hG :  $\forall g \in G, \forall g' \in G, g \sqcap g' \in G$ ) (
  u u' : Valuation  $\Omega$ )
  (h :  $\forall g \in G, u g = u' g$ ) { $\iota$  : Type*}
  [DecidableEq  $\iota$ ]
  (s : Finset  $\iota$ ) (a :  $\iota \rightarrow \Omega$ ) (ha :  $\forall i \in$ 
  s, a i  $\in G$ ) :
  u (s.sup a) = u' (s.sup a)
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1825
theorem Valuation.product_unique (v w :
  Valuation (LowerSet  $\mathbb{N}$ ))
  {u u' : Valuation (LowerSet ( $\mathbb{N} \times \mathbb{N}$ ))}
  (hu :  $\forall A B : \text{LowerSet } \mathbb{N}, u (A \times^s B) =$ 
  v A * w B)
  (hu' :  $\forall A B : \text{LowerSet } \mathbb{N}, u' (A \times^s B)$ 
  = v A * w B) : u = u'
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noncomputable def ultrafilterValuation (W
  : Ultrafilter  $\iota$ ) : Valuation (Set  $\iota$ )
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1846
theorem rect_mem_iff { $\alpha \beta$  : Type*} {W :
  Ultrafilter ( $\alpha \times \beta$ )}
  {S : Set  $\alpha$ } {T : Set  $\beta$ } :
  S  $\times^s$  T  $\in W \leftrightarrow S \in W.\text{map Prod.fst} \wedge T \in$ 
  W.map Prod.snd
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1860
theorem product_valuation_not_unique :
   $\exists u u' : \text{Valuation (Set } (\mathbb{N} \times \mathbb{N}))$ ,
  ( $\forall S T : \text{Set } \mathbb{N}, u (S \times^s T) = u' (S \times$ 
   $^s T) \wedge u \neq u'$ )
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1870
theorem fubini_fails_for_valuations :
   $\exists (v w : \text{Valuation (Set } \mathbb{N})) (u u' :$ 
  Valuation (Set ( $\mathbb{N} \times \mathbb{N}$ ))),
  ( $\forall S T : \text{Set } \mathbb{N}, u (S \times^s T) = v S * w$ 
  T)  $\wedge$ 
  ( $\forall S T : \text{Set } \mathbb{N}, u' (S \times^s T) = v S *$ 
  w T)  $\wedge u \neq u'$ 
1871

```

```

1871 (fun i => Valuation.mixCountable (
1872   u i) (hu i) (v i))
1873 = Valuation.mixCountable (fun p : Σ
1874   i, κ i => w p.1 * u p.1 p.2)
1875   (tsum_sigma_mul_eq_one w hw u hu
1876   ) (fun p => v p.1 p.2)
1877
1878 theorem Valuation.mixCountable_unique (v :
1879   Valuation Ω) :
1880   Valuation.mixCountable (fun _ : PUnit
1881     => 1) (by simp) (fun _ => v) = v
1882
1883 theorem Valuation.mixCountable_comm {ι κ :
1884   Type*}
1885   (w : ι → ℝ≥0∞) (hw : Σ' i, w i = 1) (u
1886     : κ → ℝ≥0∞) (hu : Σ' j, u j = 1)
1887   (v : ι → κ → Valuation Ω) :
1888   Valuation.mixCountable w hw (fun i =>
1889     Valuation.mixCountable u hu (v i))
1890   = Valuation.mixCountable u hu
1891     (fun j => Valuation.mixCountable
1892       w hw (fun i => v i j))
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C The Aczél Development

This appendix sketches the from-scratch construction behind the product-rule half of Theorem 30. The target shape is Aczél’s associativity theorem [Aczél 1966]: a suitably regular associative $F : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ regroups to multiplication, $G(F x y) = G(x) \cdot G(y)$ for all x, y and a strictly monotone G . `mathlib` has no associativity representation, no Hölder-style embedding of ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity).

Hypotheses. A Scale bundles: associativity of F , *divisibility* $\forall c \exists x, F x x = c$, positivity on the cone $x < F x c$, joint order-preservation, joint continuity, a unit u , and identity at $-\infty$, meaning $F x z \rightarrow x$ as $z \rightarrow -\infty$. Order-preservation was absent from our first bundle, and the additivity proof needs it. Divisibility supplies the 2^n -th roots that the dyadic construction below counts, and without it the generator cannot be written down. It is taken as a plain hypothesis in the bundle, although its interval form is a consequence of the intermediate value theorem (`exists_diag_eq`). Non-vacuity: *log-sum-exp*, $F x y = \log(e^x + e^y)$ with $u = 0$, satisfies every field (`logSumExpScale`), and for it the intended generator is globally exp.

Construction. Write $a^{(n)}$ for n right-associated copies of a under F (`Fpow`). Associativity alone makes $n \mapsto a^{(n)}$ a semigroup morphism from $(\mathbb{N}, +)$, with no commutativity needed. Commutativity is a *corollary* of the generator, not a prerequisite. One fact usually assumed is here *derived*: unboundedness of the iterates. Were they bounded,

monotone convergence would give a fixed point $FLa = L$, contradicting positivity, and unboundedness yields the Archimedean property. Divisibility supplies a tower of 2^n -th roots $u \succ u^{1/2} \succ u^{1/4} \succ \dots$ (`droot`) with the coherence law that 2^ℓ fine copies recombine one coarse mark. Counting how many copies of the n -th root fit under b (`dcount`, well defined by the Archimedean property) gives dyadic approximants $g_n(b) = \text{dcount}(b, n)/2^n$. The refinement sandwich $2N + 1 \leq N' \leq 2N + 2$, where N and N' are the counts at consecutive refinement levels n and $n + 1$, makes them monotone and Cauchy, and the generator is their limit, $g(b) = \bigsqcup_n g_n(b)$.

Results (all sorry-free, standard axioms). On the cone $\{x \mid u \leq x\}$ we prove three things. Normalization: $g(u) = 1$. Additivity: $g(F x y) = g(x) + g(y)$, by squeezing super- and sub-additive counting bounds (`dcount x + dcount y + 1 ≤ dcount(F x y) ≤ dcount x + dcount y + 3`) against a vanishing 2^{-n} error term. And *strict* monotonicity, proved without any continuity of g by a count-amplification argument: a shrinking root inserted between $x < y$ contributes a full extra block of 2^{n-M} copies, M being the refinement level of the inserted root. Exponentiating gives the multiplicative form, a strictly monotone G with $G(F x y) = G(x) G(y)$ on the cone (`exists_mul_generator`), which is the conclusion shape of Aczél’s theorem restricted to the cone (`aczelStatement_cone`). A bounded reorientation $\bar{G} = \exp(-g)$ transports it to the $[0, 1]$ -valued conjunctive orientation of Section 6.

Remaining gap. The constructed g is discontinuous at the unit (it is 0 below u), so matching the classical statement verbatim still needs the off-cone extension (a group completion) and continuity of the extension. That is genuine analysis, open here, but not vacuous: for the log-sum-exp example the global continuous generator exists (`hasOrderedGenerator_logSumExp`). Of Hölder’s classical hypotheses, the Archimedean property, which rules out elements too fine for any real number to detect and is what makes an embedding into $\mathbb{R}$ possible at all, is a theorem in this development rather than an assumption, and the interval form of divisibility follows from the intermediate value theorem. Only the embedding’s global regularity remains.